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Regional Patterns of Magnetite Authigenesis in the Appalachian Basin: Implications for the Mechanism of Late Paleozoic Remagnetization

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We report paleomagnetic and rock magnetic data from sedimentary carbonates of the Appalachian Basin that have implications for the mechanism of widespread late Paleozoic remagnetization in this rock type. Sampling was accomplished mostly along two across-basin transects, one in Devonian carbonates of New York State and the other in Ordovician and Mississippian carbonates of Tennessee and Alabama. Paleomagnetic investigations of the New York Devonian carbonates along a transect from Albany to Buffalo indicate that these rocks were completely remagnetized in Alleghenian time. Rock magnetic studies show that magnetite concentration varies in a regular fashion along the transect with a local maximum near Syracuse and decreasing gradually to the east and west. This pattern shows a striking correlation with the degree of diagenetic alteration of clay minerals in a Devonian bentonite horizon which took place during Alleghenian time. We conclude that most of the magnetite present in these carbonates is of authigenic origin and that clay mineral alteration and magnetite authigenesis were coeval, late Paleozoic events that were controlled by the same diagenetic factors. Thermoviscous remagnetization processes cannot be ruled out in this setting in light of the observed degree of thermal maturity. However, our results suggest that chemical factors played a critical role in remagnetization since they allow us to infer that magnetite authigenesis and remagnetization are about the same age. We propose a geochemical model for magnetite authigenesis wherein the iron is derived from detrital smectites during diagenetic illitization, which is triggered by the introduction of potassium-rich brines. Results from the Tennessee transect from Nashville to Chattanooga show a very different pattern. Paleomagnetic studies indicate that late Paleozoic remagnetization has affected Ordovician carbonates of the Nashville Dome. Paleozoic carbonates in the overthrust belt near Chattanooga also carry the late Paleozoic remagnetization. However, Mississippian carbonates between the dome and the overthrust belt are very weakly magnetized and show no evidence for late Paleozoic remagnetization. Samples show only a present field magnetization or, at one locality in northwestern Alabama, a dual-polarity magnetization of probable Mississippian age. Rock magnetic studies indicate that higher concentrations of magnetite are present in the remagnetized rocks of the Nashville Dome and the overthrust belt than in the nonremagnetized Mississippian carbonates. We therefore conclude that magnetite authigenesis and remagnetization are related events in this setting and that the remagnetization must be due to chemical processes. The presence of a remagnetization "shadow" in the younger rocks now exposed between the fold-thrust belt and the flexural arch is attributed to westward flow of magnetite-forming aqueous fluids through deep, sub-Carboniferous aquifers during Alleghenian time. Upward migration of the fluids through the Mississippian carbonates may have been prevented by the Devono-Mississippian Chattanooga Shale, an impermeable stratum that is present throughout the region.

Introduction

Over the past 5 years it has become well established that Paleozoic sedimentary rocks in eastern North America commonly carry a remagnetization of late Paleozoic age. Two different types of remagnetization have been observed. One type is common in red beds, where the overprint resides in hematite as a thermally distributed magnetization [e.g., *Kent and Opdyke*, 1985; *Miller and Kent*, 1986]. The red bed remagnetizations are typically syn-Alleghenian ones that partially obscure a thermally discrete high unblocking temperature magnetization of pre-Alleghenian and presumably depositional origin. The other type of remagnetization affects sedimentary carbonates and commonly resides in magnetite. Like the red bed remagnetization, the carbonate one also appears to be syn-Alleghenian in some cases where a fold test has been possible [*McCabe et al.*, 1983; *Bachtadse et al.*, 1987; *Kodama*, 1988]. The carbonate remagnetization differs from the red bed one in that it resides in a different mineral, and it generally completely obscures any older magnetization that might be present. The red bed remagnetization is the result of chemical authigenesis [*Kent and Miller*, 1987], but the origin of the carbonate remagnetization is not determined: it may be a chemical remanent magnetization due to the postdepositional development of magnetite [*McCabe et al.*, 1983, 1984], or it may be a thermoviscous remagnetization of preexisting magnetite [*Kent*, 1985].

This paper will focus on the carbonate remagnetizations that reside in magnetite and their origin. Distinguishing between chemi-

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cal, thermochemical, and thermoviscous remagnetization processes has proven to be difficult, and the mechanism of remagnetization remains controversial. Evidence for chemical remagnetization processes includes the common observation of extracted magnetite grains with diagenetic textures and compositions and the proven secondary nature of the magnetizations [*McCabe et al.*, 1983, 1984]. In addition, the remagnetization has been observed on the craton where thermal maturity indicators suggest that significant heating has never occurred and where the remagnetization is probably a result of chemical authigenesis (see *Van der Voo* [1989] for review). However, *Kent* [1985] showed that thermal activation of preexisting magnetite cannot be ruled out in settings where moderately high temperatures have been reached, as is the case in eastern upstate New York.

Early on in the research on this subject, *McCabe et al.* [1983, 1984] suggested that orogenic fluids were responsible for the carbonate remagnetizations. Fluids driven from orogenic zones through permeable sedimentary strata may have been chemically active and facilitated the precipitation of secondary magnetite in the conduit rocks. Moreover, the fluids may have been hot and therefore been agents of thermally activated remagnetization [*Dorobek*, 1984, 1989; *Leach and Rowan*, 1986; *Jackson et al.*, 1988*b*; *Miller and Kent*, 1988]. The subject of orogenic fluids has become one of considerable general interest over the past few years, and remagnetization has frequently been cited among other evidence for lateral migration of fluids during orogenic events [e.g., *Oliver*, 1986]. Although the orogenic fluids idea is an attractive one in that it explains the origin of a number of different geologic features as the result of a single cause, the evidence is still largely circumstantial. Most of what we know about orogenic fluids comes from computer simulations [*Sharp*, 1978; *Bethke*, 1985,1986; *Bethke et al.*, 1988] and studies of the presumed consequences of the fluids. These consequences include the distribution of Mississippi Valley type ore deposits and hydrocarbon accumulations [*Oliver*, 1986], authigenic feldspar development [*Hearn and Sutter*, 1985], illitization of mixed-layer illite/smectite clays [*Elliott and Aronson*, 1987], as well as remagnetization. Additional research on the causes of remagnetization is of obvious interest to paleomagnetists seeking primary Paleozoic magnetizations. Moreover, such studies are likely to make a substantial contribution to the debate on orogenic fluids.

Until recently, systematic studies of the regional pattern of Appalachian remagnetization have not been attempted. However, preliminary results of along-strike trends in the Appalachian remagnetization age [*Miller and Kent*, 1988] suggest that remagnetization occurred somewhat earlier in the southern Appalachians than the northern Appalachians, in accord with the observed south to north progression of Alleghenian deformation. In another study, variations in magnetite concentration in Devonian limestones in an east-west transect across New York State [*Jackson et al.*, 1988*b*] were found to be strongly covariant with differences in the degree of diagenetic illitization within a Devonian bentonite horizon. It was inferred that illitization and magnetite authigenesis were controlled by the same diagenetic factors and that both occurred during the late Paleozoic, in accord with a chemical remagnetization model. The results of these investigations indicate that additional regional studies will further increase our understanding of these remagnetizations, their geologic context, and their causes. This study is a continuation of the work of *Jackson et al.* [1988*b*]. We report here the results of further rock magnetic investigations of the remagnetized Paleozoic carbonates of New York State. In addition, we present the results of a similar across-strike study of carbonates of the southern Appalachian Basin in Tennessee and Alabama.

GEOLOGICAL SETTING AND SAMPLING

In the northern field area, samples were collected from sites in the Devonian Helderberg and Onondaga carbonates at regular intervals along their outcrop belt from Albany, New York to southern Ontario, just west of Buffalo, New York (Figure 1). Additional samples were taken from exposures of the Onondaga Limestone in the Hudson River Valley (Figure 1).

Since we are interested in the causes of remagnetization during Alleghenian time, the burial setting of the New York carbonates during the late Paleozoic is of interest. Physical stratigraphy and tectonic models indicate that during this time, the Devonian carbonates were in burial environments that graded from a flexural arch (Algonquin Arch) in southern Ontario, to a foreland basin in central New York State [*Quinlan and Beaumont*, 1984]. Thermal maturity indicators are in accord with tectonic models for the area. The conodont alteration index [*Epstein et al.*, 1977] and degree of clay mineral diagenesis in the Tioga Bentonite [*Johnsson*, 1986] indicate an increasing thermal maturity from west to east along the profile in New York State between Buffalo and Syracuse. Between Syracuse and Albany, the two thermal maturity indicators yield somewhat different results: conodont indices continue to increase eastward, while clay mineral diagenesis decreases slightly. The high degree of thermal maturity in central New York is generally considered to have been acquired during Alleghenian time by burial in a foreland basin setting [*Friedman and Sanders*, 1982; *Johnsson*, 1986; *Jackson et al.*, 1988*b*]. Burial temperature estimates are sensitive to the duration of burial and are also somewhat method-dependent. Typical estimates range from near 100°C in western New York to about 200°C near Albany. Apatite fission track ages indicate that the area experienced uplift and cooling during the Mesozoic Era [*Johnsson*, 1986].

In the southern field area, Paleozoic carbonates were sampled at regular intervals between Nashville and Chattanooga, Tennessee. Carbonates were sampled at five sites in the Middle Ordovician Stones River Group and the Late Ordovician Nashville Group on the Appalachian flexural arch, locally called the Nashville Dome.

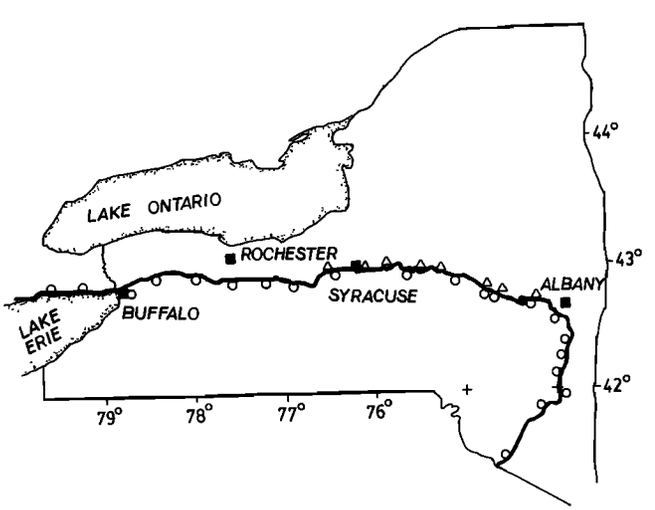


Fig. 1. Map of upstate New York showing the locations of sampling sites in the Onondaga limestone (circles) and the Helderberg limestone (triangles).

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Three sites were sampled in the Middle Mississippian Ste. Genevieve, Warsaw, and St. Louis limestones in the relatively undeformed foreland basin between the Nashville Dome and the Chattanooga Thrust. Two sites were sampled in the overthrusted fold-thrust belt near Chattanooga, one in the Late Mississippian Pennington Formation, the other in the Lower Ordovician Knox Dolomite. Figure 2 shows the locations of the sites.

One additional locality in the Mississippian trend adjacent to the Nashville Dome was sampled in greater detail. Middle Mississippian oolitic limestones of the Pride Mountain Formation were sampled at a quarry in Margerum, Alabama (Figure 2). These carbonates are somewhat unusual: they contain abundant solid bitumen which occupies ooid microporosity. The original purpose in sampling these rocks was to investigate the relationship between hydrocarbons and remagnetization, which has been reported to exist in some other settings [*McCabe et al.*, 1987; *Elmore et al.*, 1987].

The Tennessee-Alabama carbonates differ from those we sampled in New York in that their thermal maturity is very low. The conodont alteration index of 1.5 or less [*Epstein et al.*, 1977] suggests that none of the Tennessee-Alabama sites have suffered significant burial, and that burial temperatures have remained less than about 100°C since the time of deposition.

Methods

Paleomagnetic Methods

Oriented samples were subjected to stepwise alternating field (AF) and thermal demagnetization in the new paleomagnetic laboratory at Louisiana State University. Bituminous Mississippian samples from Margerum, Alabama, were demagnetized using the AF method only. Thermal demagnetizations were carried out using a three-zone furnace constructed from Lindberg components and housed in three nested mu-metal shields. The residual field is less than 50 nT in the furnace and less than 5 nT in the cooling chamber. AF demagnetization was done using a shielded coil tuned to 653 Hz and driven by a Behlman oscillator and AC power supply. Remanent magnetization was measured in a cryocooled, three-axis superconducting magnetometer manufactured by CTF Systems. Demagnetization data were plotted on orthogonal demagnetization diagrams [*Zijderveld*, 1967], and the directions of the component magnetizations were determined with principal component analysis [*Kirschvink*, 1980].

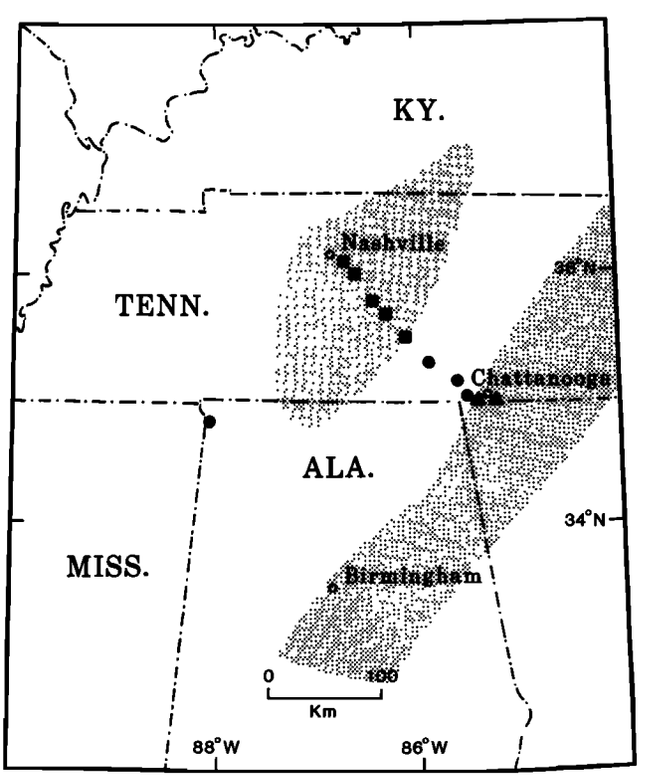


Fig. 2. Map showing the location of sampling sites for the Nashville-Chattanooga transect and for the Margerum, Alabama, locality. Symbols are as follows: solid squares, Ordovician carbonates on the Nashville Dome; solid circles, Mississippian carbonates from the Appalachian Basin; solid triangles, Paleozoic carbonates of the fold-thrust belt. Stippled areas indicate the Nashville Dome and Valley and Ridge.

Rock Magnetic Methods

In order to characterize the remanence carriers in terms of mineralogy and more particularly in terms of the concentration and size distribution of magnetite, we have measured a number of rock magnetic parameters. The overall mineralogy of the remanence-carrying fraction was characterized in terms of the acquisition and decay of isothermal remanent magnetization (IRM) [*Dunlop*, 1972; *Lowrie and Heller*, 1982]. The proportion of magnetite in the remanence-carrying fraction is qualitatively related to the parameters S_1 and S_3, defined as

$$S_1 = \frac{\text{IRM}_{-100\text{mT}}}{\text{SIRM}}$$

and

$$S_3 = \frac{\text{IRM}_{-300\text{mT}}}{\text{SIRM}}$$

[e.g., *Collinson*, 1983; *O'Reilly*, 1984; *Thompson and Oldfield*, 1986]. Relative concentrations and grain sizes of magnetite were evaluated in terms of low-field susceptibility (k) and anhysteretic susceptibility (k_A) [*Banerjee*, 1981; *King et al.*, 1982, 1983]. Relative magnetite concentration for the Tennessee-Alabama samples, which had very low bulk susceptibilities, was estimated by comparison of the IRM at 0.3 T. High-field hysteresis measurements were made on selected samples in order to evaluate further the magnetic grain sizes [*Day et al.*, 1977]. The paramagnetic/diamagnetic and ferromagnetic contributions to k ($k_P + k_F = k$) were determined from the high-field (paramagnetic) part of the hysteresis loops [e.g., *Rochette*, 1987]. Finally, coercivity spectra were obtained by measuring partial anhysteretic remanent magnetizations (pARMs) [*Jackson et al.*, 1988*a*] over alternating-field (AF) windows 10 mT wide, centered every 10 mT from 5 to 95 mT. The coercivity spectrum obtained in this way is equivalent to that obtained by differentiation of an ARM induction or AF-demagnetization curve, which is diagnostic of domain state [*Bailey and Dunlop*, 1983], and qualitatively related to grain size [*Dunlop*, 1983; *Jackson et al.*, 1988*a*].

A vibrating sample magnetometer (VSM) was used for hysteresis measurements, with loops generally extending up to 0.5 T. IRMs were imparted with either a conventional electromagnet, or an impulse magnetizer, in fields up to 1.1 T. Susceptibility was measured with a Bartington AC bridge, and remanence measurements were made on either the three-axis 2-G superconducting

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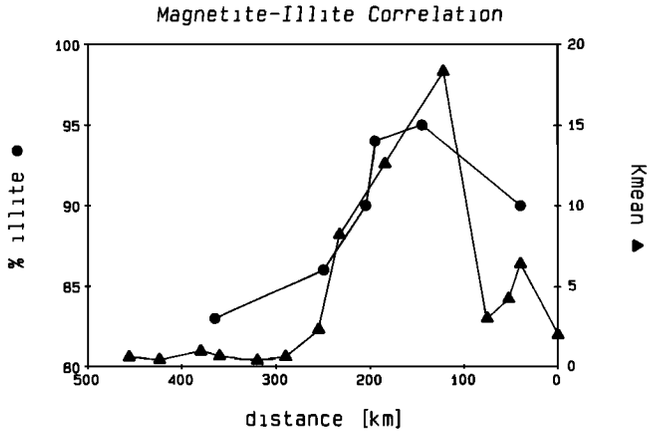


Fig. 3. Variation in anhysteretic susceptibility (triangles), reflecting magnetite concentration in the Onondaga Limestone along the transect of Figure 1. Low-field susceptibility follows the same pattern [*Jackson et al.*, 1988*b*], so k_A reflects relative magnetite content. Also shown is the percentage of illite (circles) in mixed-layer illite/smectite in the Tioga metabentonite layer within the Onondaga, [from *Johnsson* ,1986].

magnetometer at the University of Minnesota, or the CTF instrument at Louisiana State University. ARMs and pARMs were imparted in a Schonstedt static AF device, with a DC bias field of 0.1 mT. For pARMs, the DC field was switched on over the specified window of AFs by means of a relay circuit [*Jackson et al.*, 1988*a*].

RESULTS

New York Transect

Paleomagnetism. The paleomagnetism of the Devonian Helderberg Group limestones of New York State has been characterized by *Scotese et al.* [1982], who found that the unit was completely remagnetized during the Kiaman interval. *Kent* [1979, 1985] has obtained similar results for the Devonian Onondaga, Tully, and Helderberg carbonates. The Ordovician Trenton Limestone of upstate New York also carries a similar, complete remagnetization [*McCabe et al.*, 1984]. The mineral carrying the remagnetization in all of these carbonates has been shown to be magnetite. We subjected selected Onondaga and Helderberg samples from our collection to stepwise thermal demagnetization. We found that a well-defined Kiaman reversed magnetization is present all across upstate New York and in southern Ontario, in agreement with the previously reported results.

Rock Magnetism. Both k and k_A of Onondaga samples exhibit a distinct pattern of variation across the Buffalo-Albany transect (Figure 3), with uniformly low values in the western part, increasing rapidly to a peak value near Syracuse, and then declining somewhat eastward. The strong covariance between k and k_A indicates fairly uniform magnetite grain sizes, and we have therefore argued previously that the variation in each of these parameters primarily reflects differences in the concentration of magnetite [*Jackson et al.*, 1988*b*]. The ratio k/k_A is similar to those for synthetic samples with magnetite in the approximate size range 0.1-1 μm [*Banerjee et al.*, 1981].

Hysteresis properties of samples from three sites in the Hudson Valley are listed in Table 1. The J_{RS}/J_S values are in the range reported by *Day et al.* [1977] for pseudo-single domain (PSD) magnetite samples, but the H_{RC}/H_C ratios are anomalously high, reflecting a mixture of high- and low-coercivity material [e.g., *Parry*, 1982; *Nagata and Carleton*, 1987]. According to *Nagata and Carleton* [1987], H_{RC}/H_C ratios greater than 10 can be accounted for by a binary system only if the high-coercivity phase accounts for more than about 40% of the SIRM. The IRM-backfield parameters S_1 and S_3 (Table 2) indicate that the fraction of SIRM carried by high-coercivity antiferromagnetic phases such as hematite or goethite is probably less than 25-35% (= $(1-S_1)/2$) and may be as low as 5-10% (=$(1-S_3)/2$). Therefore the hysteresis properties are most likely controlled by the magnetite size distribution. The high H_{RC}/H_C ratios, combined with the PSD-like values of J_{RS}/J_S and k/k_A, point to a broad (and perhaps bimodal)

TABLE 1. Hysteresis Properties of the Knox Dolomite of East Tennessee and the Onondaga Limestone From the Hudson River Valley, New York

Sample	H_C, Oe	H_{RC}, Oe	J_S, 10^{-5} emu/g	J_{RS}, 10^{-5} emu/g	H_{RC}/H_C	J_{RS}/J_S
			*Knox Dolomite**			
BAL 1.3C	233	442	5.0	3.2	1.9	0.64
BAQ 1.1C	150	342	4.4	2.4	2.3	0.54
COP 7.12	126	394	2.7	1.4	3.1	0.54
FCP 3.1C	125	404	10.4	4.6	3.2	0.45
NM 35.1A	34	400	8.9	1.6	11.8	0.18
NM 42.3	79	422	7.1	2.9	5.3	0.41
			Onondaga Limestone			
FO1-8	64	453	37.	10.	7.1	0.27
FO2-12	56	399	48.	15.	7.2	0.31
FO3-2	49	402	43.	11.	8.2	0.25
FO5-4	49	557	18.	4.0	11.3	0.22
FO5-7	45	491	8.9	2.1	10.8	0.24
FO5-9	49	543	8.6	2.3	11.2	0.27

Abbreviations are as follows: H_C is coercivity, H_{RC} is coercivity of remanence, J_S is the saturation magnetization (corrected for paramagnetic contribution), and J_{RS} is the saturation remanence.

*Samples from the study of *Bachtadse et al.*, [1987].

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TABLE 2. IRM-Backfield Parameters

Sample	S_1	S_3
	*Knox Dolomite**	
BAL 1.3C	0.32	0.65
BAQ 1.1C	0.65	0.99
COP 7.12	0.56	0.94
FCP 3.1C	0.58	0.90
NM 35.1A	0.51	0.94
NM 42.3	0.51	0.89
	Onondaga Limestone	
FO1-8	0.49	0.92
FO2-12	0.54	0.91
FO3-2	0.54	0.94
FO5-4	0.27	0.82
FO5-7	0.28	0.82
FO5-9	0.25	0.83

*Samples from the study of *Bachtadse et al.*, [1987].

distribution of magnetite grain sizes, from SD through MD, with the former accounting for most of J_{RS} [*Nagata and Carleton*, 1987] and the latter accounting for most of J_S (i.e., most of the total volume of magnetite) [*Parry*, 1982].

The coercivity spectra of samples from the Hudson Valley and from the Syracuse area, as measured by pARM, are shown in Figure 4. The spectra exhibit very little variability, either within each area or between the two areas. Comparison with the spectra of synthetic samples [*Jackson et al.*, 1988*a*] suggests a predominance of magnetite particles in the nominal size range of one to several microns, but with a significant contribution from finer, higher-coercivity grains. This accords well with the above interpretation of the hysteresis data. The presence of larger particles (tens of microns), shown previously by the Lowrie-Fuller test [*Kent*, 1979] and by SEM imaging of extracts [*McCabe et al.*, 1983], is not well resolved in the pARM spectra. This is probably due to the relative insensitivity of anhysteretic remanences to coarse particles, but a possible alternative explanation is that the large diagenetic magnetites in these rocks have higher coercivities than synthetic magnetites of corresponding size. In either case it appears that a mixture of magnetic grain sizes is present in all samples and that the size distribution does not vary significantly across the study area.

Figure 5 shows a very strong correlation between k_F and k_P for Onondaga samples from three localities in the Hudson Valley. This indicates a nearly constant ratio of magnetite to nonremanence-carrying Fe-bearing minerals, e.g., clays and hexagonal pyrrhotite, which is present in some samples of the Helderberg Limestone [*McCabe et al.*, 1983; *Freeman*, 1986]. This could conceivably be explained in terms of either authigenic coprecipitation of magnetite and some paramagnetic mineral(s) or local derivation of Fe for magnetite formation. We acknowledge some uncertainty in this analysis due to the presence of hematite, which remains unsaturated in fields of less than 2 T and which may therefore make a substantial contribution to the "paramagnetic" susceptibility. However, the IRM backfield parameters S_1 and S_3 suggest that variations in the magnetite/hematite ratio are larger than those in k_F/k_P, which in turn suggests that the apparent paramagnetic susceptibilities are not dominantly due to hematite. Therefore the proportionality of ferromagnetic and truly paramagnetic susceptibilities is probably still valid.

Tennessee-Alabama Samples

Paleomagnetism. The paleomagnetic results from the Tennessee-Alabama sampling fall naturally into two groups which will be discussed separately: (1) results from Ordovician carbonates of the Nashville Dome, and from both early and late Paleozoic carbonates of the Tennessee fold/thrust belt, and (2) results from Mississippian carbonates from the relatively undeformed region between the flexural arch and the fold-thrust belt.

Ordovician carbonates from five sites on the Nashville Dome and Mississippian carbonates from a site near the outcrop of the Chattanooga Thrust were found to carry a well-defined characteristic magnetization with typically Kiaman reversed directions. Some samples of the Ordovician Knox Dolomite collected at one site just southeast of Chattanooga also appeared to carry a Kiaman remagnetization, in agreement with the results of *Bach-*

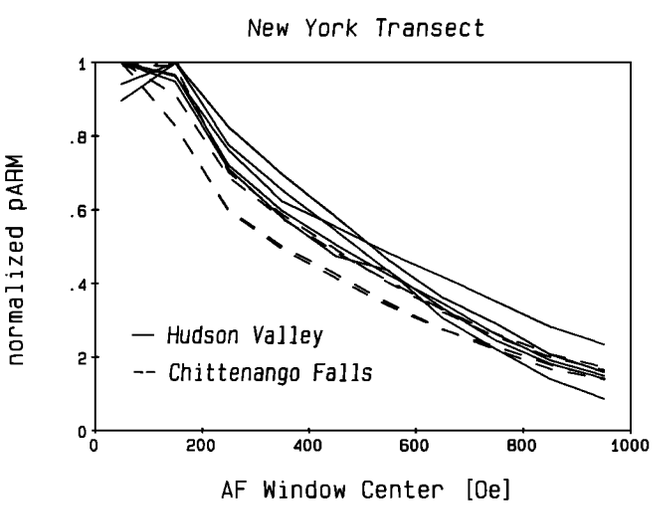


Fig. 4. Remanent coercivity spectra of Onondaga samples from the Hudson Valley and from Chittenango Falls, near Syracuse, obtained by measurements of partial anhysteretic remanent magnetization [*Jackson et al.*, 1988*a*].

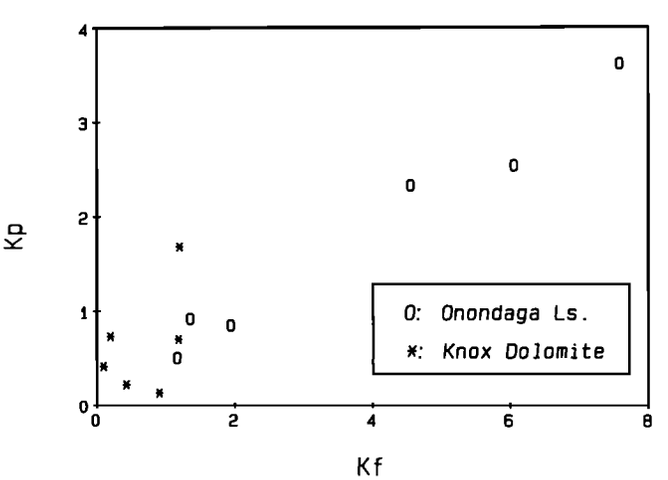


Fig. 5. Ferromagnetic (k_F) and paramagnetic (k_P) contributions to the low-field susceptibility of the Hudson Valley Onondaga and Knox Dolomite samples.

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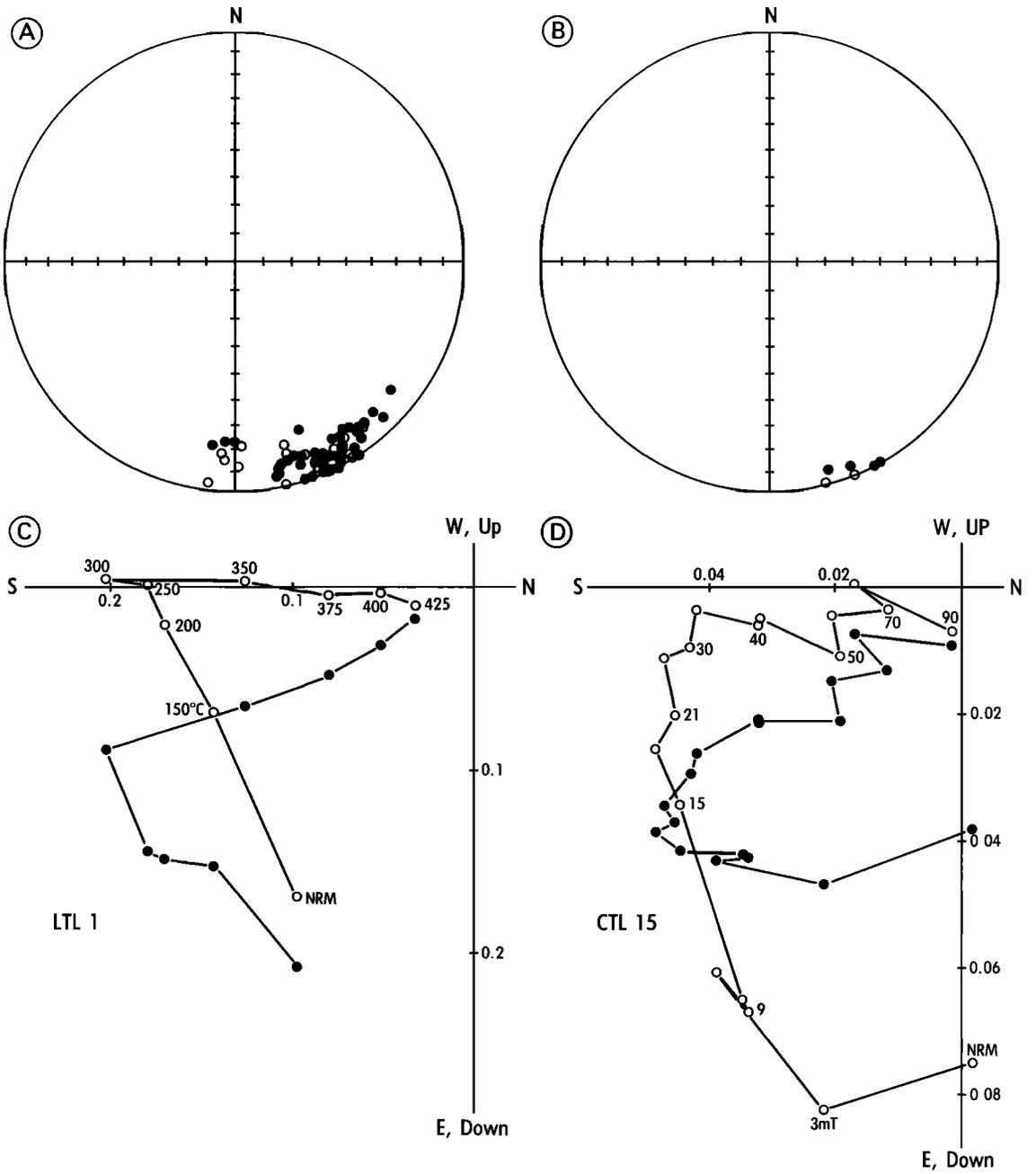


Fig. 6. (*a*) Specimen characteristic directions for Tennessee samples from the Nashville Dome and fold-thrust belt that were remagnetized during Kiaman time. (*b*) Site mean directions. (*c*) Typical thermal and (*d*) AF demagnetization data for remagnetized Tennessee specimens.

tadse et al. [1987] for the Knox of east Tennessee. However, this site was excluded from the paleomagnetic analysis because some of the samples yielded anomalous (southeast and steeply up) directions that gave rise to suspicions that an out of place block was sampled. The results of AF and thermal demagnetization were similar, although thermal treatment yielded somewhat better results (Figure 6*c* and 6*d*). In most samples treated, a magnetization aligned with the present field of probable viscous origin was removed below about 300°C or 30 mT. The characteristic magnetization was revealed by treatments between about 300°C and 500°C (Figure 6*c*) or 30 and 100 mT (Figure 6*d*). The characteristic magnetization showed good clustering at both the within and between site levels (Table 3 and Figures 6*a* and 6*b*). The mean paleomagnetic pole is located at 47.9°N, 127.9°E (K=165, A_{95}=5.2°), in good agreement with Permian poles from cratonic North America [*Van der Voo*, 1989]. These results are similar to those reported for the Cambrian Nolichucky carbonates of east Tennessee [*Gillett*, 1982] as well as those from the Knox study of *Bachtadse et al.* [1987].

Mississippian carbonates collected at three sites from the Appalachian Basin between the Nashville Dome and the Chattanooga Thrust (Figure 2) have entirely different paleomagnetic characteristics. Stepwise thermal and AF demagnetization revealed no evidence for remagnetization during the Kiaman Interval. Typical

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TABLE 3. Summary of Site Paleomagnetic Data for Tennessee Samples That Carry a Kiaman Interval Remagnetization

Site ID	Location		N/N_0	D/I, deg	k	α_{95}, deg	Pole	
	Latitude °N	Longitude °W					Latitude °N	Longitude °E
NTL	36.1	86.7	17/26	163.7/ 7.0	16.5	9.0	47.6	117.8
LTL	36.0	86.6	11/12	157.9/ -0.2	43.2	7.0	48.6	128.1
MTL	35.8	86.4	4/5	165.5/ -1.1	61.2	11.8	52.3	117.7
CTL	35.7	86.3	7/12	152.1/ 0.7	40.0	9.7	45.6	135.5
FTL	35.5	86.1	15/16	150.5/ 0.7	59.0	5.0	44.8	137.9
RTL*	35.0	85.4	9/14	157.7/ 5.0	14.9	13.8	47.1	128.4
Means				157.9/ 2.0 k=142.1 α_{95}=5.6°			47.9 K=165.3 A_{95}=5.2°	127.9

N/N_0 is the number of specimen directions used compared with the number of specimens demagnetized. D is declination, I is Inclination. K, A_{95}, k, and α_{95} are statistical parameters [*Fisher*, 1953].

*Results from overthrusted Mississippian strata near Chattanooga. Other sites are from the Ordovician carbonates of the Nashville Dome.

demagnetization diagrams are shown in Figures 7*b* and 7*c*, which show a present field magnetization as the only well-resolved component present. Specimen directions for this component as determined with principal component analysis are shown in Figure 7*a*. Some of the Mississippian samples behave during thermal demagnetization like the example given in Figure 7*b*. Although a north and down magnetization is the only well-isolated component in this specimen, demagnetization does not proceed toward the origin but moves toward either a southerly and down direction or a northerly and up direction (as in Figure 7*b*). This suggests that a dual polarity, pre-Kiaman magnetization acquired in the southern hemisphere may be present, but unfortunately, it is never resolved well in the vector diagrams. Great circle analysis cannot be used to determine this magnetization since the rocks were taken from flat-lying strata. If this poorly defined magnetization is of pre-Kiaman age as we suspect, then our results suggest that Kiaman remagnetization, so common in carbonates in eastern North America (and elsewhere), is absent in this setting. While it is possible that a partial Kiaman remagnetization has been obscured by the Cenozoic overprint, the significant point still remains that the effects of the late Paleozoic remagnetization event were much less extensive in this area than on the Nashville Dome to the west or in the overthrust belt to the east.

One additional locality along the same Mississippian trend as the one sampled in Tennessee was studied in greater detail. Some of these samples, collected at a quarry in Margerum, Alabama, did reveal a probable depositional magnetization upon demagnetization, and no evidence for Kiaman overprinting was found. We found at an early stage of the laboratory work on the Margerum collection that many of the samples with very weak magnetizations did not yield any useful information upon demagnetization. Accordingly we initiated laboratory treatment of a specimen only if the magnetization at NRM exceeded 0.025 mA/m. Even then, a large number of specimens did not demagnetize stably. Of the original collection of 161 samples (one specimen per sample), data from 87 were rejected due to either weak magnetizations or unstable behavior. Because of the presence of bitumen in the samples, thermal demagnetization was judged impractical, and only AF demagnetization was used to clean the samples.

The remaining 74 specimens demagnetized with reasonable stability, as can be seen in the demagnetization diagrams shown in Figure 8. Most of the samples have a present field magnetization as the only well-resolved component present, as in the example shown in Figure 8*c*. However, some samples have a well-defined magnetization with southeast and moderate down (Figure 8*a*), or northwest and moderate up directions (Figure 8*b*). Other samples have two components, one aligned with the present field, and the other with the ancient magnetization (Figure 8*d*). Figure 9*b* shows the directions of all the magnetization components judged to be ancient ones, and Figure 9*a* shows all other components obtained from linear demagnetization trends. The ancient magnetization has a mean (normal) direction of D=325.9, I=-30.8 (N=17, k=6.4, α_{95}=15.4°). The paleomagnetic pole is at 29.2°N, 130.0°E, in agreement with results of proven Mississippian age from elsewhere in North America [*Irving and Strong*, 1984; *Kent and Opdyke*, 1985]. Because of the small number of ancient directions obtained, the lack of a field test to bracket its age, and the high rejection rate, this is obviously not a determination of high quality. The significance of this result lies in the fact that a believably Mississippian magnetization is preserved in these carbonates and that no evidence for Kiaman overprinting was found. This result tends to confirm our inference, described earlier, that Kiaman overprinting did not affect the Mississippian carbonates of Tennessee west of Chattanooga and east of the Nashville Dome. The lack of Kiaman remagnetization in this setting may be an important observation with respect to the ultimate origin of the remagnetization that affected rocks to the east and to the west.

Finally, we note that the presence of a resolvable early magnetization in these bituminous rocks is somewhat surprising in the light of recent studies which suggest that the presence of hydrocarbons is related to the development of diagenetic magnetite in some settings [*McCabe et al.*, 1987; *Elmore et al.*, 1987].

Rock Magnetism. The coercivities and blocking temperatures observed during demagnetization of our Tennessee and Alabama samples suggest that magnetite may be the dominant carrier mineral in all of the rock units we studied. Additional rock magnetic tests were conducted to test this inference. During stepwise IRM acquisition almost all of the samples reached saturation, or near

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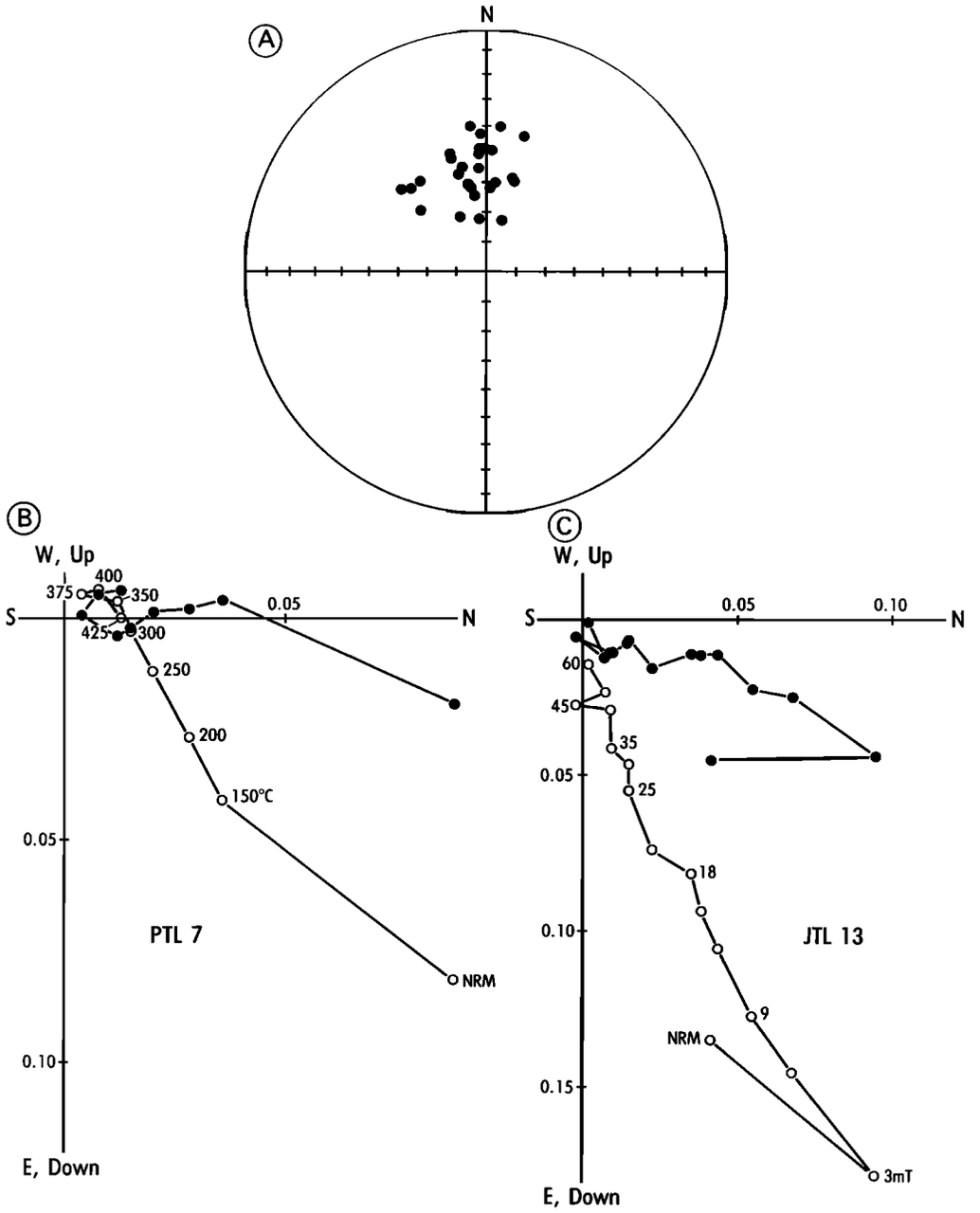


Fig. 7. (*a*) Specimen directions for Tennessee Mississippian samples that do not show any evidence for Kiaman Interval remagnetization. (*b*) Thermal and (*c*) AF demagnetization data for Tennessee Mississippian samples.

saturation by 0.2-0.3 T, indicating that high-coercivity phases are not present to any great extent in most samples and that magnetite is likely to be the dominant magnetizable phase (Figure 10). To investigate the intrinsic blocking temperatures of the carriers, representative samples from Tennessee were given an IRM at 1.1T along the -V sample axis, and an IRM at 0.2 T along the +V axis. The samples were then thermally demagnetized [*Channell et al.*, 1982]. Stepwise thermal demagnetization of this composite IRM shows that magnetization is removed continuously from low-coercivity grains up to 500-575°C (Figure 11) in both remagnetized and nonremagnetized samples from Tennessee, thus confirming that magnetite is present in significant quantities. A small number of samples (e.g., PTL 6, Figure 11) also contain some hematite.

Since thermal demagnetization of the bituminous samples from Margerum, Alabama, is not practical, magnetic isolates were prepared for direct observation of the carrier grains. This work was difficult, since the concentrations of magnetic phases were very low in most of the samples. However, the magnetic isolates from some of the samples were observed to be dominated by detrital magnetite grains. Figure 12 shows scanning electron micrographs of magnetite grains (in polished grain mount) extracted from Margerum samples. Qualitative energy dispersive X ray analysis indicated that only iron and titanium were present as major elements, and the SEMs clearly show high-temperature oxidation-exsolution textures, indicating that the grains are detrital magnetites with exsolved ilmenite. The presence of detrital magnetite

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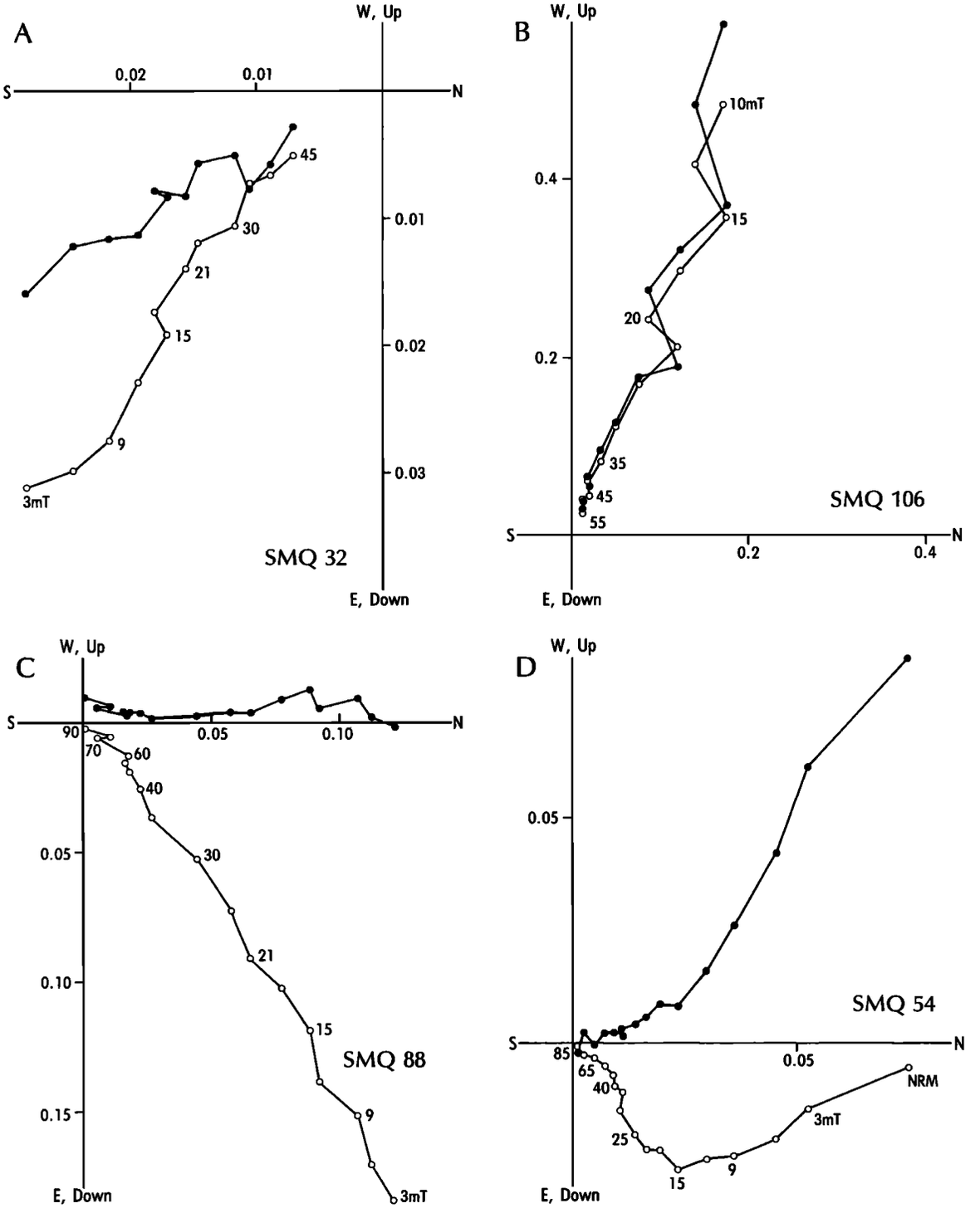


Fig. 8. Typical AF demagnetization data for Mississippian carbonate samples from Margerum, Alabama. (*a*) A specimen with a pre-Kiaman reversed magnetization. (*b*) A specimen with a pre-Kiaman normal polarity magnetization. (*c*) A specimen containing only a present field magnetization. (*d*) A specimen with both a present field and an ancient reversed magnetization.

in the Margerum samples is consistent with the apparently depositional magnetization found in some of the samples from that locality. No magnetic isolates have yet been prepared from our Tennessee samples, but we intend to begin a systematic petrographic investigation of the Tennessee carbonates in the near future, including studies of both diagenetic history and magnetic carrier morphology and composition.

Both k and k_A exhibit considerable variability along the Nashville-Chattanooga transect, but unlike the New York samples, there is little covariance of k and k_A. The low-field susceptibilities do not show any strong spatial pattern; however, k_A is observed to be relatively high in samples from the Nashville Dome and from the fold-thrust belt and low in the Mississippian samples from the intervening basin. Because the bulk susceptibilities are negative in

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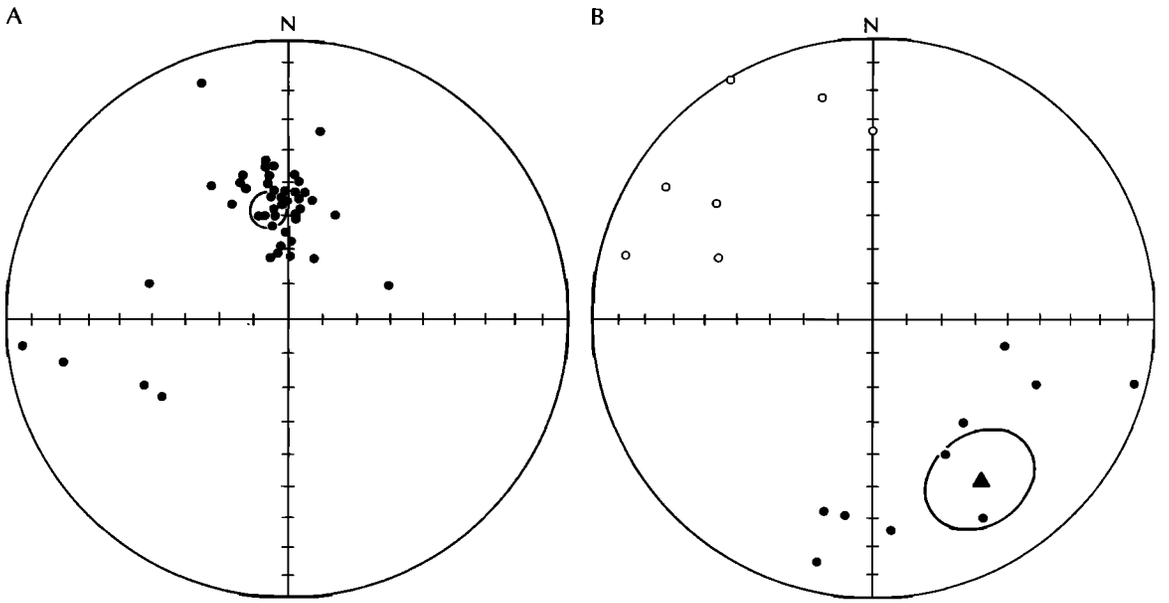


Fig. 9. Specimen directions from linear demagnetization trajectories for Mississippian samples from Margerum, Alabama. (*a*) Directions not used for determination of the mean characteristic direction. The α_{95} of the mean of all directions is shown. (*b*) Specimen characteristic directions interpreted to be a depositional magnetization. Mean direction is shown as a triangle, surrounded by its α_{95}.

many cases, it is clear that the magnetite concentration is quite low; further, because of the obvious importance of nonferromagnetic contributions to k, no quantitative inferences can be made from k/k_A regarding magnetite concentrations or grain sizes. IRM intensity follows the same general trend as k_A. In order to estimate magnetite concentration for comparison along the transect, samples were subjected to stepwise IRM acquisition up to 1.1 T (Figure 10). Estimated relative magnetite concentration was compared using the IRM at 0.3 T (Figure 13), using only data from samples in which saturation at 0.3 T was demonstrated. The results indicate that the Mississippian samples from Alabama and Tennessee between the flexural arch and the fold-thrust belt, which do not show evidence for Kiaman Interval remagnetization, are uniformly magnetite-poor with saturation IRMs ranging from several milliamperes per meter to about 50 mA/m with a mode in the 0-10 mA/m range (Figure 13*a*). On the other hand, remagnetized samples from the Nashville Dome and from the fold-thrust belt have IRMs that range from about 10 mA/m to about 160 mA/m (Figure 13*b*). All of these observations are above the mode IRM from the nonremagnetized samples. Clearly, the Kiaman-remagnetized

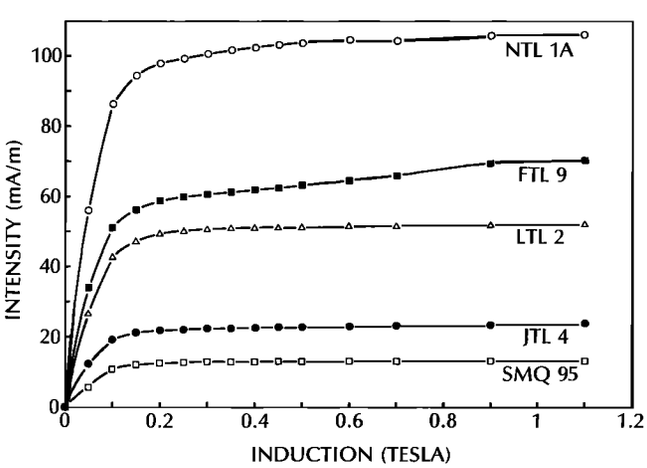


Fig. 10. Typical IRM acquisitions of Tennessee and Alabama samples. Each of the samples reaches saturation or near-saturation by 0.2-0.3 T, indicating that the magnetizable grains are characterized by low coercivities. The two lower curves are from nonremagnetized Mississippian samples; the upper three curves are from samples that were remagnetized in the late Paleozoic.

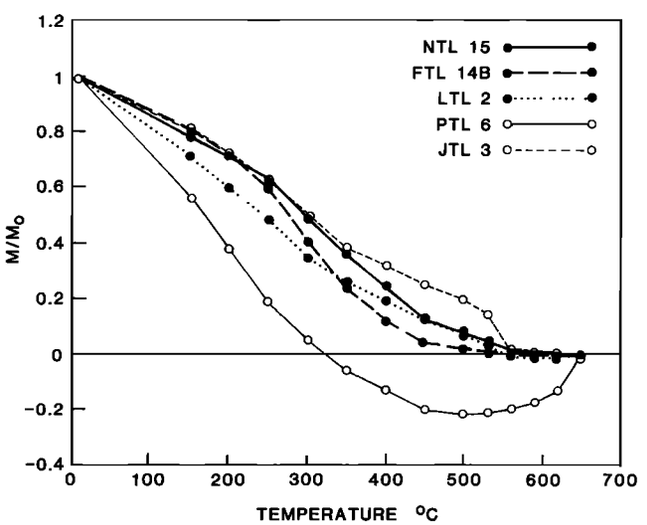


Fig. 11. Thermal demagnetization of composite IRM using the method of *Channell et al.* [1982] for Tennessee samples that were remagnetized during the late Paleozoic (solid circles) and Mississippian samples that show no clear evidence for Paleozoic remagnetization (open circles). Low-coercivity carriers have a positive magnetization, and high-coercivity grains have a negative one. The results indicate that all samples contain magnetite. One anomalous sample (PTL 6) also contains some hematite. Only a few hematite-bearing samples were found in our collection.

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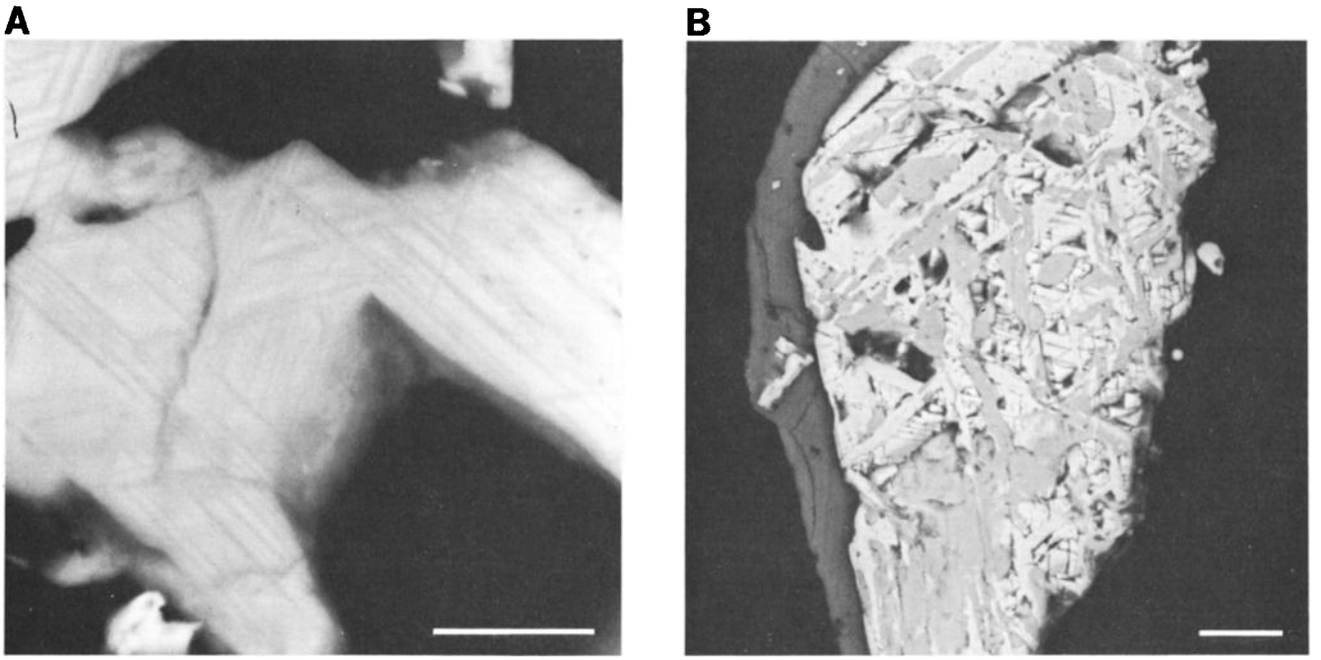


Fig. 12. Scanning electron micrographs of two magnetite grains, in polished grain mount, extracted from the Mississippian limestones collected at Margerum, Alabama. (*a*) The grain clearly shows the high-temperature oxidation-exsolution of ilmenite, indicating a detrital origin of this magnetite grain. Scale bar indicates 10 μm. (*b*) The grain shows a high-temperature oxidation texture. The blebby "exsolution" is incipient martitization. Scale bar indicates 20 μm.

samples have, on average, more magnetite than samples that lack evidence for remagnetization. This observation suggests that the formation of new magnetite is related to the remagnetization phenomenon in this setting.

Hysteresis measurements for the Knox Dolomite of the Tennessee fold-thrust belt (samples of *Bachtadse et al.* [1987]) are summarized in Table 1. Again the results are quite different from those obtained for the Onondaga Limestone. J_{RS}/J_S is generally higher and more variable; H_{RC}/H_C is lower (although still quite high for the corresponding values of J_{RS}/J_S) and much more variable. IRM backfield measurements (Table 2) indicate that high-coercivity antiferromagnetic material (i.e., hematite and/or goethite) is not of much importance except in one sample (BAL 1.3C). Thus, as in the Onondaga samples, the rather large H_{RC}/H_C ratios are probably due to the presence of a broad range of magnetite grain sizes, volumetrically dominated by MD particles. The J_{RS}/J_S ratios, appearing to indicate a predominance of SD-like particles, may be somewhat erroneously high due to undersaturation of hematite/goethite at the peak fields used (generally 0.5 T).

The ratio k_F/k_P is quite variable for the Tennessee samples (Table 4). If most of the magnetite present is of postdepositional origin, the variability in this parameter might suggest that either (1) Fe was brought in from somewhere else by mineralizing fluids and magnetite was precipitated inhomogeneously over the region, or (2) the magnetite formed dominantly by diagenetic alteration of an inhomogeneously distributed preexisting Fe-bearing phase, perhaps pyrite, which has a very low susceptibility [*Rochette*, 1987]. There is once again an uncertainty due to the probable presence of hematite, but in this case the fairly constant IRM backfield parameters suggest that variations in hematite content are probably not the cause of the variable k_F/k_P.

Partial ARM curves for the Knox samples (Figure 14*a*) are virtually identical to those of the Onondaga samples (Figure 4). A window-by-window comparison of the mean normalized pARMs for the two sets of samples is shown in Figure 14*b*; the correlation is nearly perfect. It is worth pointing out here that a very similar coercivity spectrum is also apparent in the remagnetized Trenton Limestone [*McCabe et al.*, 1984]; differentiation of the AF demagnetization curve shown by *McElhinny and Opdyke* [1973] for ARM yields a spectrum indistinguishable from those of the Knox and Onondaga. The very strong similarities in this respect exhibited by these three units indicate that this coercivity spectral pattern may be a fundamental and diagnostic property of samples with dominantly authigenic magnetite.

Farther west, all along the Nashville-Chattanooga transect, the coercivity spectra are systematically shifted to somewhat higher values, suggesting an overall westward fining of magnetite grain sizes (Figure 14*a*). This departure from the coercivity spectra of both the Onondaga Limestone of New York and the Knox Dolomite of the fold-thrust belt might be related to the fact that authigenic magnetite concentrations in the Nashville-Chattanooga setting are lower. One might be seeing, for example, the effect of a relatively greater contribution of depositional magnetite to the coercivity spectra in Tennessee west of the fold-thrust belt.

DISCUSSION

West of the Hudson Valley in New York State, the magnetite concentration in the Onondaga and Helderberg carbonates as estimated by k and k_A [*Jackson et al.*, 1988*b*] correlates directly with the percentage of illite in mixed-layer illite/smectite in the Tioga Bentonite [*Johnsson*, 1986]. Because percent illite is considered to be influenced by diagenetic temperatures [*Hower*, 1981; *Johnsson*,

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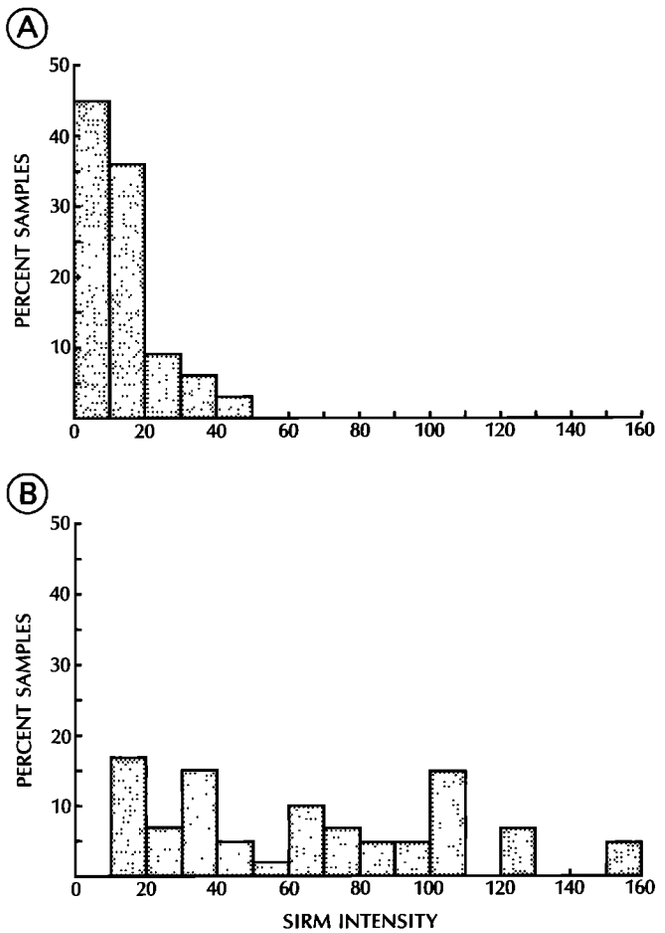


Fig. 13. Histogram of IRM magnetization at 0.3 T for (*a*) Mississippian samples from Tennessee and Alabama that show no evidence for Kiaman Interval remagnetization and (*b*) Tennessee samples from sites that were remagnetized during the Kiaman Interval.

1986], we have previously argued that magnetite authigenesis in the Onondaga was temperature-dependent [*Jackson et al.*, 1988*b*]. However, in order to explain both the pattern of percent illite, with a maximum near Syracuse, and that of conodont color alteration indices (CAI [*Epstein et al.*, 1977]), which increase monotonically eastward, we inferred that paleotemperatures were not the sole control on either percent illite or magnetite concentration. We therefore proposed a diagenetic model [*Jackson et al.*, 1988*b*], in which chemical factors related to the introduction of orogenic fluids during Alleghenian time play an important role in triggering both illitization and magnetite authigenesis.

The data presented in this paper suggest a possible geochemical mechanism for magnetite authigenesis in New York State wherein the introduction of exotic, potassium-rich brines triggers both magnetite authigenesis and illitization of detrital clays within the limestones. The pertinent observations are as follows: (1) magnetite concentration is closely related to the extent to which illitization has occurred, (2) there is a direct relationship between k_F and k_P, which might indicate that paramagnetic clay mineral concentration and magnetite concentration are covariant parameters within the limestones, and (3) there is evidence for orogenic fluids that were potassium rich, as indicated by reports of Alleghenian ages for authigenic K-feldspars and K-bentonites in the basin [*Hearn and Sutter*, 1985; *Elliott and Aronson*, 1987]. Since illitization of detrital smectite requires potassium, and the process may release iron [*Boles and Franks*, 1979], it is possible that illitization, K-feldspar development, and magnetite authigenesis are genetically related processes that were triggered by the introduction of potassium-rich fluids (J.S. Hanor, personal communication, 1988). Although this geochemical model is speculative, we believe it is testable with further field and experimental studies, which are underway.

Several workers have previously inferred that illitization [*Johnsson*, 1986], time of maximum burial [*Friedman and Sanders*, 1982; *Johnsson*, 1986], and magnetite authigenesis [*Jackson et al.*, 1988*b*] in New York State were Alleghenian

TABLE 4. Magnetic Susceptibility and Anhysteretic Susceptibility Data From the Knox Dolomite and Onondaga Limestone

Sample	k	k_P	k_F	k_A	k_P/k_A
			*Knox Dolomite**		
BAL 1.3C	0.50	0.41	0.09	22.60	0.004
BAQ 1.1C	0.64	0.22	0.42	17.10	0.025
COP 7.12	0.92	0.73	0.19	9.43	0.020
FCP 3.1C	1.88	0.70	1.18	25.60	0.046
NM 35.1A	1.03	0.13	0.90	4.84	0.19
NM 42.3	2.88	1.68	1.20	8.67	0.14
			Onondaga Limestone		
FO1-8	6.89	2.33	4.56	46.60	0.10
FO2-12	11.20	3.61	7.59	57.90	0.13
FO3-2	8.59	2.54	6.06	49.30	0.12
FO5-4	2.79	0.85	1.94	13.10	0.15
FO5-7	2.27	0.92	1.35	7.35	0.18
FO5-9	1.66	0.50	1.16	7.57	0.15

Susceptibility values are in 10^{-6} emu Oe^{-1} cm^{-3}. Abbreviations are as follows: k is the low-field susceptibility and k_A the anhysteretic susceptibility; k_P and k_F are the paramagnetic and ferromagnetic contributions to the low-field susceptibility respectively.

*Samples from the study of *Bachtadse et al.*, [1987].

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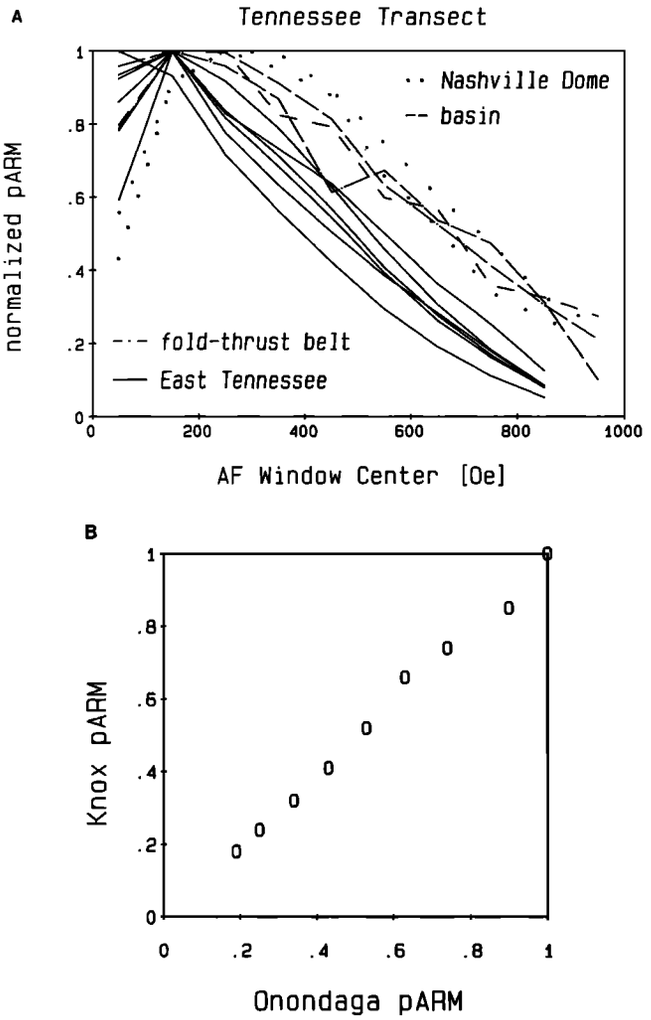


Fig. 14. (*a*) Remanent coercivity spectra of samples from the Tennessee transect and of Knox samples from east Tennessee, obtained by measurements of partial anhysteretic remanent magnetization. (*b*) Correlation plot of mean normalized pARMs for each AF window, for the Knox and Onondaga samples.

events. Significantly, an unpublished K-Ar age from the bentonite horizon within the Onondaga studied by *Johnsson* [1986] also indicates that illitization there occurred during Alleghenian time (R. Reynolds, personal communication, 1989). Since remagnetization of the Onondaga is also an Alleghenian event [*Kent*, 1985] and our data suggest a temporal link between illitization and magnetite authigenesis, we infer that magnetite authigenesis and remagnetization are about the same age and that chemical processes must have played a critical role in the remagnetization. However, because of the high thermal maturity in evidence over much of eastern upstate New York, thermally activated remagnetization cannot be dismissed lightly [*Kent*, 1985].

A problem with the thermoviscous remagnetization model is that a remagnetization due to burial and uplift would not be expected to result in a remagnetization of late Paleozoic age in New York State, since apatite fission track ages indicate that uplift and cooling of the area occurred during Mesozoic time [*Johnsson*, 1986]. Although possibly Cretaceous remagnetizations have been isolated from Paleozoic carbonates in Vermont [*Tucker and Kent*, 1988], no remagnetizations associated with Mesozoic unroofing have been reported for any Paleozoic carbonates in New York State. Even so, thermally activated late Paleozoic remagnetization might have occurred in this setting if the heat was supplied advectively by the episodic migration of very hot fluids during the Alleghenian orogeny.

Evidence for episodic migration of high-temperature, high-pressure orogenic fluids has been reported for the Helderberg carbonates of the central Appalachians [*Dorobek*, 1984, 1989]. The evidence includes the observation of homogenization temperatures in secondary fluid inclusions of 200° to 300°C, which greatly exceed paleotemperatures expected from thermal maturity data and known overburden thickness [*Dorobek*, 1984, 1989]. Using a simple conductive cooling model, *Dorobek* [1989] shows that if hot fluid migration events were episodic, the thermal anomalies due to them could decay rapidly enough to have had a minimal effect on commonly used thermal maturity indicators, which record integrated time-temperature rather than simply paleotemperature. If this situation applies to the carbonates of New York State, then thermoviscous processes could reasonably be inferred to have played a role in the observed late Paleozoic remagnetization despite the fact that unroofing occurred much later. If the same fluids were involved in both magnetite authigenesis and heat transport, then the remagnetization could have been due to both chemical and thermal magnetization processes. Additional studies of secondary fluid inclusions in a number of Appalachian carbonate environments are needed to help resolve this issue. *Dorobek's* [1984, 1989] results appear to be of immense importance in explaining the causes of the carbonate remagnetizations, but confirmation of them needs to be sought: fluid inclusion studies are difficult, and a number of factors can result in homogenization temperatures that are much higher than the actual filling temperature.

The pattern of late Paleozoic paleotemperatures and magnetite authigenesis in the southern Appalachian Basin of Tennessee-Alabama is in marked contrast with the situation in New York State. The observed conodont alteration index of about 1.5 for both the Nashville Dome carbonates and the Mississippian carbonates of the basin to the east [*Epstein et al.*, 1977] suggests maximum burial temperatures of 100°C or less, much too low for thermally activated remagnetization due to burial and uplift. We emphasize that the lack of remagnetization in the Mississippian strata is therefore not attributable to a difference in burial temperatures. Moreover, the higher average magnetite concentration in the remagnetized Ordovician carbonates of the Nashville Dome compared with the nonremagnetized Mississippian carbonates of the basin indicates that magnetite authigenesis probably played a dominant role in the remagnetization.

The observed patterns of magnetite concentration and remagnetization in the southern Appalachian Basin are consistent with a diagenetic remagnetization model in which fluids that migrated laterally from the orogenic zone toward the craton are the agents responsible for remagnetization. We propose that the fluid pathways west of the fold-thrust belt were confined mostly to the lower Paleozoic portion of the section and that therefore remagnetization was confined to the early Paleozoic rocks. Independent evidence for an Alleghenian episode of migration of K-rich brines exists in the form of late Paleozoic K-Ar ages from Ordovician K-bentonites within the same units we sampled [*Elliott and Aronson*, 1987]. We suggest that the introduction of exotic potassium may have been responsible for both magnetite authigenesis and illitization of the bentonites, in accord with the geochemical model proposed above. The Mississippian carbonates may have escaped remagnetization because they were hydrologically isolated

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from the main fluid conduits beneath them. A likely barrier to the vertical migration of magnetite-forming fluids is the Chattanooga Shale, an impermeable stratum of Devono-Mississippian age that is present throughout the study area. Though chemical processes are most likely responsible for remagnetization on the Nashville Dome, we cannot completely exclude thermoviscous effects. Again, if the fluids were hot and they moved in episodic pulses, thermal effects might have had a role in the remagnetization without having much affect on thermal maturity indicators [*Dorobek*, 1984, 1989]. However, since the observed thermal maturity is so low on the Nashville Dome and likely sources of hot fluids are so distant, we regard thermoviscous effects to have been unimportant in this setting.

CONCLUSIONS

Regional patterns of remagnetization and magnetite concentration are quite different in two across-strike studies of Appalachian Basin carbonates. In New York State, remagnetization has affected Devonian carbonates all across the state, and magnetite concentration varies in a pattern reflecting differences in the late Paleozoic diagenetic environment as inferred from studies of the degree of illitization in mixed-layer illite/smectite. However, in Tennessee and Alabama, higher magnetite concentrations are found in remagnetized carbonates of the fold-thrust belt and the Nashville Dome than are found in a remagnetization "shadow" in the intervening basin. We attribute this contrast between the northern and southern basin environments to the much greater depth of burial that existed in New York State during the late Paleozoic. The New York results reflect remagnetization in a deep basinal setting, whereas the Tennessee-Alabama results reflect near-surface conditions at the time of remagnetization. Remagnetization in both settings can be explained in terms of lateral migration of potassium-bearing fluids from the orogenic zone toward the craton. We suggest that fluid conduits were laterally continuous strata that plunged to deep burial levels during Alleghenian time: these conduits were the sites of magnetite authigenesis and remagnetization. Carboniferous strata at shallow levels of the Tennessee-Alabama foredeep escaped significant remagnetization because they were not along fluid conduits. The Chattanooga Shale may have prevented the upward migration of the remagnetizing fluids into the late Paleozoic carbonates. Most of the strata of the overthrusted Valley and Ridge were subject to remagnetization due to the enhanced permeability to remagnetizing fluids resulting from pervasive faulting.

The results of this study suggest that additional investigation of regional trends in rock magnetic and paleomagnetic parameters might eventually lead to a much better understanding of remagnetization phenomena, their geologic context, and their applications to other areas of earth science. Such studies are underway at the present time.

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